EMMANUEL G. MAMINTA

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EDUCATION

University of the Philippines Diliman, Quezon City, Philippines
Ph.D. in Artificial Intelligence, Sep 2022–Present; expected completion 2027
Adviser: Prof. Rowel O. Atienza; Ubiquitous Computing Laboratory
M.Eng. in Artificial Intelligence, Sep 2022–Jul 2025
B.S. in Electrical Engineering, with honors, Aug 2018–Jul 2022

RESEARCH INTERESTS

Embodied AI, multimodal AI, and vision-language models for retail applications (RetailVision and Physical Retail AI); fine-grained grocery product retrieval and embodied agents in virtual retail environments. Additional areas of study: hand–object interaction, egocentric and exocentric computer vision, robotics, and world models.

RESEARCH EXPERIENCE

Doctoral Researcher, Ubiquitous Computing Laboratory Sep 2022–Present
University of the Philippines Diliman, Quezon City, Philippines

- Co-authored a zero-shot evaluation of **190 open-source vision-language models** for grocery product retrieval, studying pre-training data, architecture, and input resolution (ICPR 2026).
- Led research on an agentic framework for embodied AI in virtual retail; co-authored Sari Sandbox, a simulation with over 250 interactive grocery items for benchmarking embodied agents.
- Investigated open-vocabulary object detection in vision-language models through research supported by Samsung Research & Development Institute Philippines.

PUBLICATIONS

Emmanuel G. Maminta and Rowel O. Atienza. 2026. A Transferable Variant Signal That Grocery Retrieval Cannot Reach. Asian Conference on Computer Vision (ACCV 2026), **Main Conference**. **Accepted for oral presentation**, Osaka, Japan, December 2026.

Emmanuel G. Maminta and Rowel O. Atienza. 2026. Late Interaction Ranking for Effective Closed-Catalog Grocery Product Retrieval. European Conference on Computer Vision (ECCV 2026), RetailVision Workshop. Presented in Malmö, Sweden, September 2026.

Emmanuel G. Maminta and Rowel O. Atienza. 2026. What Matters for Grocery Product Retrieval with Open Source Vision Language Models. International Conference on Pattern Recognition (ICPR 2026). Presented in Lyon, France, August 2026. **RRPR reproducibility badge**.

Janika Deborah Gajo, Gerarld Paul Merales, Jerome Escarcha, Brenden Ashley Molina, Gian Nartea, **Emmanuel G. Maminta**, Juan Carlos Roldan, and Rowel O. Atienza. 2025. Sari Sandbox: A Virtual Retail Store Environment for Embodied AI Agents. IEEE/CVF International Conference on Computer Vision (ICCV 2025), RetailVision Workshop. Presented in Honolulu, Hawai‘i, USA, October 2025.

HONORS AND AWARDS

Most Active Contributor Award, Microsoft Research Asia Aug 2026
Awarded during the AI Summer School at the National University of Singapore. Submitted the most contributions to the Global AI Value Challenge during the summer school.

Best of ICCV 2025, Voxel51 feature Nov 2025
Sari Sandbox was featured in Voxel51’s Best of ICCV 2025 research event.

PROFESSIONAL EXPERIENCE

AI Developer, OpenText Philippines Jan 2025–Present

- Build enterprise AI products, agentic systems, and LLMOps/MLOps infrastructure; apply parameter-efficient fine-tuning (PEFT) and harness engineering.
- Developed an in-house small language model to automate proprietary coding tasks for teams in the Philippines and the United States.
- Engineered knowledge graph-guided retrieval-augmented generation (RAG) with a unified data model for internal applications.

Data Scientist, Bank of the Philippine Islands Nov 2023–Dec 2024

- Engineered personalized product recommendations for high-net-worth clients and Jobs-to-Be-Done-based unsupervised clustering to identify customer segments.

Software Engineer, Samsung Research & Development Institute Philippines
Apr–Oct 2023

- Built a proprietary library for efficient, scalable deep learning experimentation.

OPEN-SOURCE CONTRIBUTIONS

- **OpenMPR, Initiator.** Started an open-source project sharing recipes and evaluation resources for fine-grained grocery product retrieval.
- **stable-worldmodel, Contributor.** Contributed to a platform for reproducible world model research and evaluation.
- **GEPA-Lite, Creator.** Released a lightweight implementation of Genetic-Pareto prompt optimization under the MIT License.
- **GEPA, Contributor.** Added the AnyMaths adapter for prompt optimization on mathematical word problems and LLM reasoning.

TEACHING AND RESEARCH COMMUNITY

Cohere Labs Open Science Community, Member Sep 2026–Present
Focus on multimodal AI and world models.

Teaching contributions, University of the Philippines Diliman 2022
Co-taught EEE 111 (Introduction to Programming and Computation) and EEE 121 (Data Structures and Algorithms) with Prof. Jhoanna Rhodette I. Pedrasa.

CERTIFICATIONS AND TRAINING

NVIDIA-Certified Professional: Agentic AI
NVIDIA-Certified Associate: Generative AI LLMs
Oxford Machine Learning Summer School, Jun–Jul 2024; enrolled in representation learning and generative AI courses.

TECHNICAL SKILLS

Modeling: Python, PyTorch, Transformers (Hugging Face), OpenCLIP, PEFT, OpenCV.
Agent systems and evaluation: DSPy, LangGraph, LlamaIndex, FastAPI, MLflow, Weights & Biases, Langfuse.
Inference and compute: vLLM, NVIDIA Triton, ONNX, Slurm, NVIDIA Jetson, AWS, Azure, Google Cloud.